

Table 1: National Air Passenger Taxes Used as Events

Country	Tax	Announced	Effective	Rate (EUR per departing passenger)	Transfers exempt	Treated airports	Control countries
Netherlands	Vliegbelasting (ticket tax)	2007-02-07	2008-07-01	11.25 / 45 (within 2,500 km or EU / other)	yes, 24 h	5	13
Ireland	Air Travel Tax	2008-10-14	2009-03-30	2 / 10 (up to 300 km / other)	yes, 6 h	9	13
Germany	Luftverkehrsteuer	2010-06-07	2011-01-01	8 / 25 / 45 (three distance bands)	yes, 12-24 h	34	11
Austria	Flugabgabe	2010-10-23	2011-04-01	8 / 20 / 35 (three distance bands)	yes, 24 h	6	12
Norway	Flypassasjeravgift	2015-12-14	2016-06-01	NOK 80 ($\approx$ 8.6), flat	yes, 24 h	50	14
Sweden	Flygskatt	2017-06-08	2018-04-01	SEK 60 / 250 / 400 ($\approx$ 6.2 / 25.8 / 41.2)	yes, 24 h	38	13
United Kingdom	Air Passenger Duty, doubled	2006-12-06	2007-02-01	GBP 5 / 20 increase ($\approx$ 7.4 / 29.6)	yes, 24 h	57	15
Denmark	Passageravgift, extended to domestic	1997-05-06	1998-01-01	DKK 75 ($\approx$ 10.1) on domestic departures	yes	13	16
Malta	Departure tax, doubled	2004-11-24	2005-08-01	Lm 10 increase ($\approx$ 23.3), flat	yes, journeys starting in Malta	2	13

Notes: Announcement is the first public government decision (coalition agreement, cabinet or budget statement, parliamentary decision), read from primary sources; effective is the first departure date taxed. Rates are the statutory amounts at introduction (or the change, for the UK, Denmark and Malta), converted at the rate of the event month where not in euro. All nine taxes exempt connecting passengers (within the stated window, on a single ticket). Treated airports are those with scheduled departures in the calendar year before the announcement; control countries are the other coded European countries (Table 2 notes) with no tax event of their own within three years before the announcement and two years after the effective date. The Netherlands is analysed separately (Table 10) because the tax was set to zero after twelve months. Sources: national legal gazettes and budget documents (coded in `taxes_west_europe.csv`, `taxes_nordic_rest.csv`, `announcement_dates_verified.csv`).

Table 2: *Summary Statistics: Airports in the Year before the Announcement*

	Airports	Seats (thousand)	Flights	Stage length (km)	GACI
<i>Taxed country</i>					
Mean	214	1,993	15,909	717	0.841
Median		181	3,056	507	0.712
<i>Border, 50–300 km</i>					
Mean	298	1,689	14,118	673	0.844
Median		149	2,432	533	0.704
<i>Control countries</i>					
Mean	2,282	1,530	12,360	671	0.812
Median		127	2,021	506	0.684

Notes: Airport-level values in the calendar year before each event’s announcement, pooled over the nine events (an airport appears once per event in which it is in the sample). Seats and flights are scheduled departures (OAG schedules); stage length is seat-kilometres per seat; GACI is the Global Airport Connectivity Index of that year. Border airports are airports of other countries within 50–300 km of a treated airport; airports within 50 km are excluded from all groups. Coded countries: Germany, Austria, Netherlands, Belgium, Luxembourg, France, Ireland, United Kingdom, Switzerland, Sweden, Norway, Denmark, Finland, Iceland, Portugal, Spain, Greece, Hungary, Croatia, Malta (Italy excluded throughout because its surcharge taxes transfer passengers and has undated changes).

Table 3: *Mean Log Seats Twelve Months before and after the Effective Date*

	Raw ln(seats)		Residualised	
	Before	After	Before	After
Taxed country	9.863	9.796	0.034	−0.034
Border airports	9.989	10.089	−0.015	0.015
Control countries	9.655	9.687	−0.013	0.013

Notes: Monthly airport observations in the twelve months before and after the effective date of the nine tax increases (stacked by event, Netherlands included). Residualised values are deviations from airport-by-calendar-month-by-event means, so they remove each airport’s level and seasonality; the difference in their before/after change between taxed and control airports is the raw difference-in-differences. Border airports within 150 km in this table.

Table 4: *Parallel-Trend Test: Event-Time Coefficients Relative to the Year before the Announcement*

	Pre, years −3 [A−36, A−25]	Pre, years −2 [A−24, A−13]	Post, year 1 [E, E+11]	Post, year 2 [E+12, E+23]
Pooled, 8 events	−0.051 (0.045)	−0.031 (0.021)	−0.124*** (0.032)	−0.153*** (0.040)
Netherlands 2008	0.016 (0.226)	−0.048 (0.043)	−0.103 (0.124)	
Ireland 2009	−0.119 (0.078)	−0.109* (0.061)	−0.109 (0.078)	
Germany 2011	0.016 (0.061)	−0.009 (0.038)	−0.180*** (0.061)	
Austria 2011	−0.124** (0.051)	−0.116*** (0.028)	−0.124** (0.061)	
Norway 2016	0.026 (0.034)	0.048** (0.022)	−0.135*** (0.038)	
Sweden 2018	0.074 (0.060)	0.011 (0.028)	−0.159*** (0.055)	
United Kingdom 2007	−0.211*** (0.046)	−0.082*** (0.027)	−0.010 (0.027)	
Denmark 1998	−	−0.131*** (0.045)	−0.296*** (0.079)	
Malta 2005	0.038 (0.052)	0.047* (0.025)	−0.633* (0.365)	

Notes: Stacked difference-in-differences on monthly $\ln(\text{seats})$ with airport-by-calendar-month-by-event and month-by-event fixed effects. Event time runs from the announcement (A) for the pre-period and from the effective date (E) for the post-period; the months between announcement and effective date are dropped (anticipation donut). Bins are twelve calendar months each, so seasonality cannot enter the comparison; reference bin [A−12, A−1]. Border airports (50–300 km) enter as a separate group (not shown). Pooled rows: standard errors clustered by country (33 clusters). Event rows: one treated country, standard errors clustered by airport, for reference only; Denmark has no third pre-year in the data. The Netherlands row refers to its 2008 introduction. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 5: *Effect of a Ticket Tax on Scheduled Seats in the First Year*

Weights	(1) Equal	(2) Seats	(3) Equal	(4) Seats
Taxed country $\times$ post	−0.103** (0.046)	−0.041* (0.021)		
small airports (bottom tercile)			−0.240*** (0.066)	−0.242*** (0.067)
mid tercile			−0.016 (0.060)	−0.021 (0.050)
large airports (top tercile)			−0.048 (0.051)	−0.040* (0.023)
Border airports $\times$ post	0.009 (0.027)	−0.030* (0.016)	0.010 (0.027)	−0.030* (0.016)
<i>N</i>	112,876	112,306	112,876	112,306
Countries	33	31	33	31

Notes: Dependent variable $\ln(\text{monthly departing seats})$. Stacked difference-in-differences over the eight tax increases of Table 1 (Netherlands separate), pre-period the 36 months before the announcement, post-period the first twelve months from the effective date, anticipation months dropped. Fixed effects: airport-by-calendar-month-by-event and month-by-event. Size terciles are defined within each treated country by seats in the year before the announcement (median seats 30 thousand, 168 thousand and 2.0 million; the top tercile holds 94% of the treated countries' seats). Columns (2) and (4) weight observations by pre-announcement-year seats, so they measure the effect on national seat totals. Border airports: other countries' airports within 50–300 km of a treated airport. Standard errors clustered by country in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 6: Robustness of the First-Year Seat Effect

Specification	Taxed $\times$ post	(s.e.)	N	Countries
Effective-date window ± 12 months	-0.115***	(0.024)	66,818	27
Effective-date window ± 18 months	-0.130***	(0.030)	93,915	27
Effective-date window ± 24 months	-0.128***	(0.036)	116,250	27
Effective-date window ± 36 months	-0.130**	(0.052)	158,773	27
Without the crisis-year events (NLD 2008, IRL 2009)	-0.111***	(0.025)	51,894	27
Announcement donut, 2 pre-years	-0.128***	(0.040)	111,063	27
Donut, airport pre-trends removed	-0.153***	(0.036)	107,904	27
Donut, 3 pre-years, verified announcement dates (main)	-0.108**	(0.050)	95,270	33
<i>Dose: tax in EUR per departing passenger (airport-specific, from distance bands and pre-year stage length)</i>				
Dose $\times$ post, per EUR	-0.0118***	(0.0037)	108,031	33
Dose $\times$ post, with treated $\times$ post included	-0.0037	(0.0051)	108,031	33
Dose $\times$ pre [A-36,A-25]	-0.0042	(0.0028)		
Dose $\times$ pre [A-24,A-13]	-0.0020	(0.0015)		

Notes: Each row is the treated-by-post coefficient of a separate stacked regression of $\ln(\text{seats})$ with the fixed effects of Table 5; border airports included as a separate group (not shown). Window rows use the effective date, nine events (Netherlands included), symmetric windows. The crisis row drops the two events whose first tax year overlaps the 2008–09 recession. Donut rows drop the months between announcement and effective date and take the pre-announcement years as reference; the pre-trend row removes each airport’s linear trend fitted on the 24 pre-announcement months. The first four donut rows use the announcement dates as coded before verification (Austria 2011-01, Malta 2004-12); the main row uses the verified dates (2010-10, 2004-11). Dose rows interact event time with the airport’s predicted tax per passenger (one distance band per airport, from its pre-year mean stage length; standard deviation 4.9 EUR among treated airports). Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 7: Margins of Adjustment by Airport Size: Frequencies, Aircraft Size and Service Loss

	Seats ln	Flights ln	Seats per flight ln	Service loss (0/1)
Small airports $\times$ post	-0.240*** (0.066)	-0.237*** (0.080)	-0.002 (0.039)	0.044** (0.021)
Mid tercile $\times$ post	-0.016 (0.060)	-0.052 (0.065)	0.036 (0.022)	-0.006 (0.007)
Large airports $\times$ post	-0.048 (0.051)	-0.073* (0.042)	0.025** (0.011)	-0.003 (0.007)
Border airports $\times$ post	0.010 (0.027)	0.019 (0.028)	-0.009 (0.013)	0.015 (0.009)
N	112,876	112,876	112,876	105,659

Notes: Same stacks and fixed effects as Table 5, column (3). Service loss is a linear probability model on the months at risk (airports with seats twelve months earlier; months with no scheduled departure count as zero seats); the mean monthly loss rate among treated small airports before the announcement is used as the base in the text. Seats per flight rises at large airports while flights fall, so hubs keep seats by up-gauging; small airports lose frequencies and whole months of service. Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 8: *Network Position of Taxed Airports: Connectivity Index and Centralities*

	ln GACI	ln Destinations	ln Eigenvector	ln Betweenness
Small airports $\times$ post	−0.023*** (0.005)	−0.135** (0.062)	−0.319*** (0.082)	−0.930** (0.406)
Mid tercile $\times$ post	−0.022*** (0.003)	−0.106*** (0.023)	−0.131** (0.049)	−0.253** (0.095)
Large airports $\times$ post	−0.003 (0.015)	−0.034 (0.054)	−0.131** (0.053)	−0.162 (0.109)
Border airports $\times$ post	−0.002 (0.005)	0.015 (0.033)	−0.100 (0.099)	0.067 (0.162)
All treated airports $\times$ post	−0.016** (0.006)	−0.089** (0.039)	−0.186*** (0.046)	−0.427** (0.165)
Hubs only (≥ 1 million seats)	−0.026** (0.010)	—	−0.138 (0.082)	−0.175 (0.116)
Seat-weighted, all airports	−0.023*** (0.007)	—	−0.074* (0.040)	−0.088*** (0.031)
<i>N</i> (annual)	13,569			

Notes: Annual airport outcomes from the GACI panel: the connectivity index (GACI), the number of destinations (degree), eigenvector centrality and $\ln(1 + 10^4 \times \text{normalised betweenness})$. Size-tercile rows: stacks of the three calendar years before the announcement and the first effective year, airport-by-event and year-by-event fixed effects (Table 5 sample). Hub and seat-weighted rows: effective-date stacks of two pre- and two post-years, nine events. GACI is normalised each year, so the coefficients are changes in relative network position. Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 9: *Synthetic Difference-in-Differences by Event: National Seat Totals*

Event	Donors	SDID	Placebo p	Synthetic control	DiD	Pre-fit RMSE	Largest donor weights
NLD 2008-07-01	13	-0.076	0.31	-0.076	-0.034	0.021	BEL 0.13, DEU 0.12, ESP 0.12, GRC 0.10
IRL 2009-03-30	13	-0.159	0.00	-0.149	-0.111	0.031	BEL 0.18, AUT 0.14, HRV 0.13, CHE 0.11
DEU 2011-01-01	11	-0.005	1.00	-0.019	-0.027	0.014	CHE 0.16, PRT 0.13, GRC 0.13, HUN 0.12
AUT 2011-04-01	12	-0.038	0.83	-0.045	-0.015	0.018	LUX 0.17, ESP 0.15, PRT 0.15, SWE 0.12
NOR 2016-06-01	14	-0.155	0.07	-0.128	-0.225	0.024	SWE 0.29, MLT 0.16, DEU 0.14, CHE 0.13
SWE 2018-04-01	13	-0.009	1.00	-0.015	-0.108	0.021	FIN 0.18, BEL 0.17, DEU 0.14, CHE 0.12
GBR 2007-02-01	15	-0.016	0.80	-0.039	-0.007	0.011	ESP 0.11, IRL 0.09, AUT 0.09, PRT 0.08
DNK 1998-01-01	16	-0.099	0.25	-0.023	-0.100	0.004	ESP 0.09, HRV 0.08, NLD 0.07, FIN 0.07
MLT 2005-08-01	13	-0.207	0.00	-0.249	-0.186	0.028	IRL 0.22, PRT 0.18, ESP 0.17, DEU 0.12

Notes: Country-by-month $\ln(\text{seats})$ of all airports of the country, de-seasonalised with country-by-calendar-month means over the pre-announcement window; pre-period the 36 months before the announcement, post-period the first twelve effective months, anticipation months dropped. SDID follows Arkhangelsky et al. (2021): unit weights on the simplex with the paper's ridge penalty, time weights, and an intercept; synthetic control uses no intercept and no time weights; DiD uses equal donor weights. Donors are the coded European countries without a tax event in the window. Placebo p is the share of donors whose own SDID estimate (treating that donor, with the actual treated country removed) is at least as large in absolute value. Malta has two airports. No significance stars: inference is by placebo only.

Table 10: *The Dutch Episode: Introduction (July 2008) and Removal (July 2009)*

	Tax in force 2008-07 to 2009-06	After removal 2009-07 to 2010-12
Dutch airports	−0.092** (0.042)	−0.023 (0.039)
Border airports, 50–150 km	0.165*** (0.052)	0.149* (0.079)
Border airports, 150–300 km	0.137 (0.169)	−0.067 (0.259)
<i>N</i>	16,687	
Countries	17	

Notes: Single-event stacked regression of $\ln(\text{seats})$ with the fixed effects of Table 5; reference period the 36 months before the February 2007 coalition agreement that announced the tax, anticipation months dropped. The tax was set to zero from 1 July 2009 and repealed in December 2009. The 50 km ring (Eindhoven, Weeze-adjacent airports) is estimated separately and is based on very few airports. Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 11: *Cross-Border Leakage: Seats at Foreign Airports near the Taxed Country*

	(1) Distance rings	(2) Border airport type
Taxed country $\times$ post	−0.108** (0.050)	−0.108** (0.050)
Border 0–50 km $\times$ post	−0.202* (0.100)	
Border 50–150 km $\times$ post	−0.019 (0.042)	
Border 150–300 km $\times$ post	−0.002 (0.030)	−0.002 (0.030)
Border hubs (≥ 5 million seats), ≤ 150 km $\times$ post		0.000 (0.025)
Border secondary airports, ≤ 150 km $\times$ post		−0.037 (0.047)
<i>N</i>	95,270	95,270
Countries	33	33

Notes: Specification of Table 5 with the border group split by great-circle distance to the nearest treated airport (column 1) or, for airports within 150 km, by size (column 2). Eight events (Netherlands in Table 10). The 0–50 km ring contains few airports and shows pre-trends in the event study; it is excluded from the other tables. Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 12: First-Year Accounting: Passengers, CO₂ and Revenue by Event

Event	Seats before (m)	Effect, year 1	Passengers lost (m)	CO ₂ avoided (kt)	Revenue (m EUR)	EUR per t CO ₂
NLD 2008-07-01	32.0	-0.103	2.5	926	1,002	1,082
IRL 2009-03-30	20.3	-0.109	1.7	196	145	744
DEU 2011-01-01	129.4	-0.180	17.0	3,825	1,112	291
AUT 2011-04-01	17.5	-0.124	1.6	239	99	415
NOR 2016-06-01	39.6	-0.135	4.0	359	238	663
SWE 2018-04-01	29.9	-0.159	3.5	423	127	300
GBR 2007-02-01	141.3	-0.010	1.1	292	1,946	6,671
DNK 1998-01-01	14.8	-0.296	3.0	433	26	60
MLT 2005-08-01	1.6	-0.633	0.6	88	16	178

Notes: Illustrative accounting that applies each event's own first-year airport-level coefficient (Table 4, event rows; equal airport weights) to the treated country's pre-announcement-year seats and departing-flight CO₂ (LTO plus cruise, from the emissions panel). Passengers assume a load factor of 0.80. Revenue is the predicted tax per passenger (distance band at the airport's mean stage length) times remaining passengers. Because the airport-level coefficient over-weights small airports, these figures bound the national effect from above; the seat-weighted estimate of Table 5 is about one third as large. EUR per tonne compares revenue with avoided emissions and is not a welfare measure.